Self-Singed Certificate 기반의 HTTPS server

1) HTTPS Server 구성

Step 1. 개인키 생성

#cd /etc/pki/tls/certs

#openssl genrsa –out www.open.net.key 2048

```
[root@open certs]# openssl genrsa -out open.net.key 2048
Generating RSA private key, 2048 bit long modulus
...........................+++
...............+++
e is 65537 (0x10001)
```

Step 2. CSR 생성

#openssl req –new –key www.open.net.key -out www.open.net.csr

*Common Name : www.open.net

```
[root@open certs]# openssl req -new -key open.net.key -out open.net.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:KR
State or Province Name (full name) []:Seoul
Locality Name (eg, city) [Default City]:Seoul
Organization Name (eg, company) [Default Company Ltd]:OpenSouce Corp.
Organizational Unit Name (eg, section) []:Security&System
Common Name (eg, your name or your server's hostname) []:www.open.net
Email Address []:admin@open.net

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
```

#ls –l /

Step 3. 개인키와 CSR을 이용해 인증서 생성

#openssl x509 –in www.open.net.csr –out www.open.net.crt –req –signkey www.open.net.key –days 365

```
[root@open certs]# openssl x509 -in open.net.csr -out open.net.crt -req -signkey open.net.key -days 365
Signature ok
subject=/C=KR/ST=Seoul/L=Seoul/O=OpenSouce Corp./OU=Security&System/CN=www.open.net/emailAddress=admin@open.net
Getting Private key
```

Step 4. 자세 서명 인증서 설치 및 설정을 위한 mod_ssl 패키지 설치

#dnf install mod_ssl –y

#rpm –q mod_ssl

Step 5. httpd-ssl.conf 파일에 인증서 경로 설정iptab

vi /etc/httpd/conf.d/ssl.conf

SSLCertificateFile **/etc/pki/tls/certs/www.open.net.crt**

SSLCertificateKeyFile **/etc/pki/tls/certs/www.open.net.key**

```
# Server Certificate:
# Point SSLCertificateFile at a PEM encoded certificate.  If
# the certificate is encrypted, then you will be prompted for a
# pass phrase.  Note that a kill -HUP will prompt again.  A new
# certificate can be generated using the genkey(1) command.
SSLCertificateFile /etc/pki/tls/certs/open.net.crt

#   Server Private Key:
#   If the key is not combined with the certificate, use this
#   directive to point at the key file.  Keep in mind that if
#   you've both a RSA and a DSA private key you can configure
#   both in parallel (to also allow the use of DSA ciphers, etc.)
SSLCertificateKeyFile /etc/pki/tls/certs/open.net.key
```

Step 6. 방화벽 설정

#firewall-cmd --add-service=https

#firewall-cmd --reload

#firewall-cmd --list-all

Step 7. 서비스 활성화

#systemctl restart httpd

2) HTTPS Client

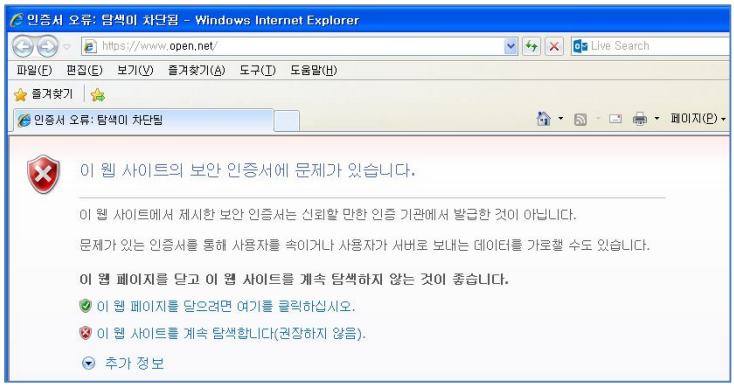
Step 1. Hosts.txt파일에 도메인등록

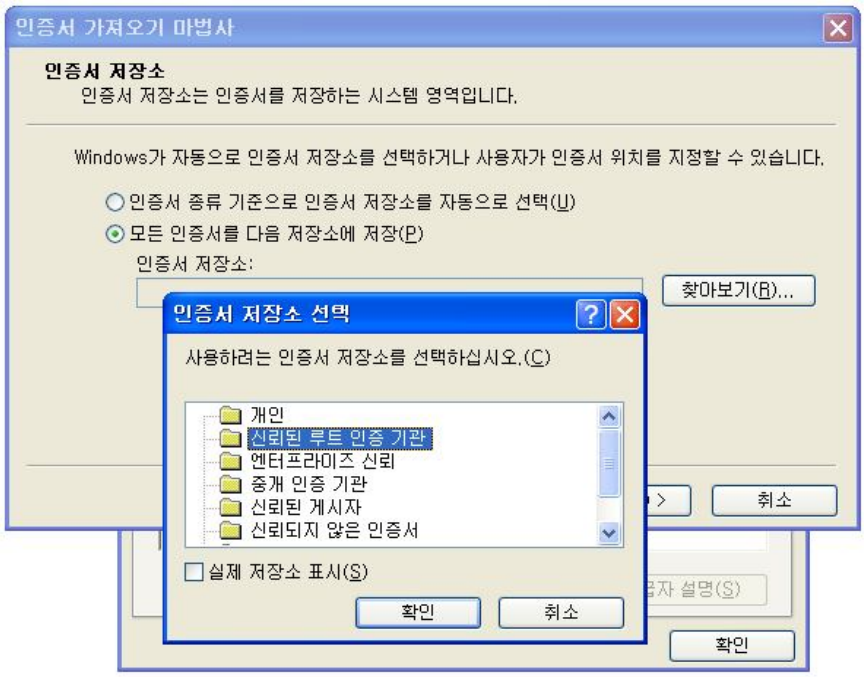
C:\>windows>system32>drivers>etc>

- Host파일 메모장이나 워드패드로 열기 도메인 명 지정

192.168.10.40 www.open.net

Step 2. https://www.open.net 접속



Step 3. 인증서 설치

'신뢰된 루트 인증기관'에 인증서 저장